

What Logiko Is

Logiko is a browser game that teaches logic by letting players build formal proofs step by step (like in natural deduction).

You get premises, a goal, and rules such as Modus Ponens, Conjunction, etc.

You win by correctly proving the goal from the premises.

How It's Built

- Frontend (UI): HTML, CSS, and JavaScript (or a framework like React).
 - Displays proof lines, rules, and user input areas.
 - Logic Engine: A program that checks each proof step.
 - Stores formulas as trees (ASTs).
 - Validates rules (e.g., checks if Modus Ponens was used correctly).
 - Parser: Converts formulas like $A \rightarrow B$ into internal data the program can check.
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Alternate Ways to Build It

- Different Languages:
 - TypeScript/React(probably react for my project) for better structure and UI.
 - Rust or OCaml compiled to WebAssembly for speed and reliability.
 - Prolog or Python (with Z3) for automated proof checking or hint generation.
 - Platforms:
 - Pure web app, or a mobile version built with Flutter or React Native.
 - Server-based version that saves progress or supports multiplayer puzzles.
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How It Could Be Improved

Easy Upgrades

- Add undo/redo and save progress.
- Better keyboard shortcuts and error messages.
- Step-by-step hints that show the next logical move.
- Increase visual effectiveness

Medium Upgrades

- Stronger formula parser with clear syntax errors.
- Adaptive difficulty based on how often a player gets stuck.

Advanced Upgrades

- Use a Rust/WASM core for fast rule checking.
 - Integrate a real proof system like Lean or Coq for full correctness.
 - Add AI-based hints or automatic puzzle generation.
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